Dr. Shivam Agarwal

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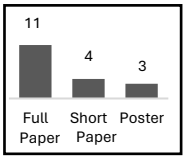
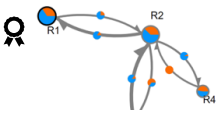
Projects



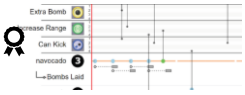
I led multiple projects, resulting in **18 peer-reviewed publications** at prestigious academic forums. Key highlights are below, categorized into three themes. For full details, visit my website.

Complex Group Behavior of AI Agents

- Designed an award-winning linked-views analytics approach that helped 63 AI expert teams in the *Flatland* 2020 competition discover and improve AI scheduling strategies. Published in *IEEE PacificVis* conference(DOI).



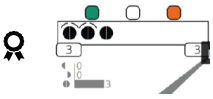
- Innovated a timeline-based method to analyze AI agents' collaborative and competitive behavior in *Pommerman*, evaluated by twenty experts. Published in *Computer Graphics Forum* journal (DOI) , this work catalyzed three subsequent research endeavors.



Dynamic Overlapping Sets

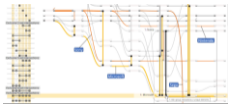
- Invented a state-of-the-art technique to analyze dynamic overlapping sets, demonstrated in three scenarios: researcher expertise, software evolution, and multi-label classification. Published in Computer Graphics Forum (DOI) with 19 citations, showcasing its high impact.

- Developed a novel layered graph approach to analyze set overlaps, demonstrated throughtwo scenarios -- researchers' field of interest and software developer activity evolution. Published in *VMV* conference (DOI) and won the best paper award.



Complex Entity Interactions

- Single-handedly invented a state-of-the-art visualization to analyze entity interactions with changing group memberships, such as companies evolving through partnerships, acquisitions, and product/service offerings. Published in *EuroVis* conference (DOI).



- Collaborated in the design of a novel visualization approach, helping researchers to analyze and improve the triage training of medical teams at mass casualty events. Published in *Computer Graphics Forum* journal (DOI).

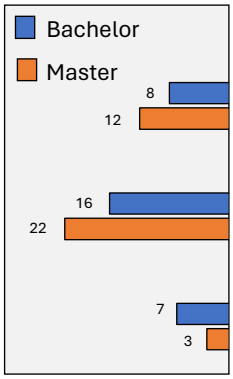
Awards

- **NeurIPS 2020's *Flatland* competition:** First prize in community contributions for visualization of multi-agent cooperative behavior.
- **Best paper award**: State-of-the-art visualization for dynamic overlapping sets using layered graphs.
- **NeurIPS 2019's *Pommerman* competition:** Community award for analyzing group dynamics.

Technical Skills

- **Languages and Frameworks:** Python, SQL, Vue.js, Flask, Web (HTML, JavaScript, and CSS).
- **Tools and Libraries:** D3.js, Tableau, Power BI, Computational Notebooks (Observable, Jupyter), Git, Scikit-learn.
- **Others:** Empirical evaluation and testing, Data visualization, Explainable AI, Machine Learning.

Supervision and Management



Supervise and manage student activities in academia.

Theses: Supervised **Bachelor's** (3 months) and **Master's** (6 months) theses on data visualization topics.

Projects: Designed and supervised student projects focused on building prototypes and data analysis.

Courses and Seminars: Managed courses and designed seminars on challenging topics.

Other Skills

- **Research and Development:** Proposed and led research projects and collaborated with international researchers.
- **Team Management:** Supervised and managed diverse research teams and designed effective strategies for inter- and intra-team interactions.
- **Collaborative and Optimistic:** Skilled in collaborating with diverse teams, maintaining an optimistic attitude, and fostering a healthy work environment.

Work Experience

Research Associate | *University of Bamberg, Germany* — *2022 - Present*

- Teaching and supervising 30+ students from diverse backgrounds with varying levels of skills.
- Developing a state-of-the-art technique that would address the current unsolved challenge of analyzing datasets containing overlapping categories, numerical, and other types of attributes.

Research Associate | *University of Duisburg-Essen,Germany* — *2017 – 2022*

- Taught 200+ students about data visualization and AI through exercise sessions, seminars, and project groups.
- Supervised 8+ students to publish and present their thesis/project contributions at reputed academic venues.
- Led eight research projects to build state-of-the-art visual analysis methods.

Software Engineer | *IIIT Bangalore, India* — *2015 – 2017*

- Spearheaded a Qt-based interface with interactive visualizations for INCOIS's geo-temporal data on water salinity, temperature, and speed at varying depths in the Indian Ocean, improving the understanding of periodic changes and anomaly detection.

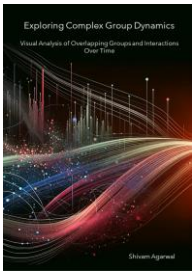
Systems Engineer | *Infosys Ltd.* — *2011 – 2012*

- Automated the application process for customers of Royal Bank of Scotland.

Books

Exploring Complex Group Dynamics: Visual Analysis of Overlapping Groups and Interactions Over Time (DOI).

- This book is a published version of my PhD dissertation.
- For summary, watch a four-minute video: URL



Education

2017 – 2023	**PhD** ***magna cum laude*** (Visual Analytics)	University of Duisburg-Essen, Germany
2013 – 2015	**Masters** (Information Technology)	IIIT Bangalore, India
2007 – 2011	**Bachelors** (Computer Science)	United College of Engineering & Research, India.

Miscellaneous

- **Languages:** English, Hindi, German (A1), and Sanskrit (Learning).
- **Hobbies:** Cooking and creating digital story books.

Volunteering and Other Activities

- **Academic Volunteering:** Volunteered as a program committee member *(IEEE VIS* – short paper), student volunteer (*ACM CHI Play*, *IEEE VIS*, and *EuroVis*), and external reviewer (*CHI, CHI Play, IEEE VIS*).
- **VisInBharat**: Co-founded a visualization community for researchers, practitioners, and students.
- **Online Tutoring:** Tutored six underprivileged students by teaching high-school mathematics in individual sessions for a period of 3 months each.

A word cloud visualization of keywords from my research articles (size encodes frequency).